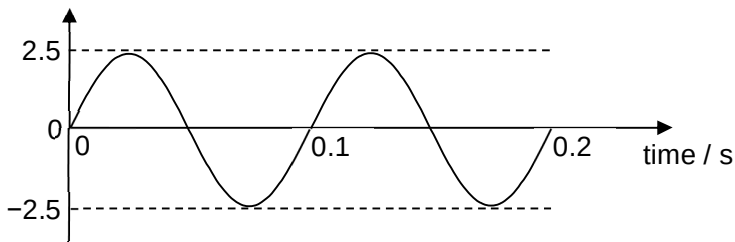


4	(a)	(Simple harmonic motion is the motion of a particle about a fixed point such that its) acceleration is proportional to displacement (from a fixed point) and always directed towards a fixed point / opposite in direction to the displacement	B1 B1
	(b)	existence of peak amplitude	C1
	(c)	microwave cooking [B1] (cook food using microwave of frequency close to the resonating frequency (of the water molecules in food)) or radio receiver [B1] (receive radio waves when the receiver (circuit) is tuned to the frequency of the radio waves to be received) or magnetic resonance imaging [B1] (generate image for medical diagnosis using radio waves of frequencies close to the resonating frequency (of the atomic nuclei))	
	(d)	max speed $v_0 = \omega x_0 = 2\pi f x_0 = 2\pi \times 10.0 \times 2.5 \times 10^{-3}$ $= 0.157 \text{ m s}^{-1}$	C1 A1
	(e)	displacement / mm  shape (sin or cos or any sinusoidal shape) [M1]	

		constant period 0.1 s and constant amplitude 2.5 mm [A1]
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5	(a)	$E = I(P + r)$	
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